TASK REPORT

Screening Task 3:

Surrogate Modelling of a Binary Distillation Column Using DWSIM and Machine Learning

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TABLE OF CONTENT :

Section No.	Topic	Page No.
1	**Problem Statement**	3
2	**DWSIM Flowsheet & Model Description**	3
3	**Input and Output Variables Used**	4
4	**Dataset Generation Strategy**	**3**
5	**Range of Operating Conditions Considered**	5
6	**Data Preprocessing Steps**	6
7	**ML Methods Implemented**	6
8	**Model Training and Validation Procedure**	**6**
9	**Performance Comparison of Models**	**7**
10	**Discussion on Physical Consistency and Robustness**	**8**
11	**Final Conclusion**	**8**

1. Problem Statement

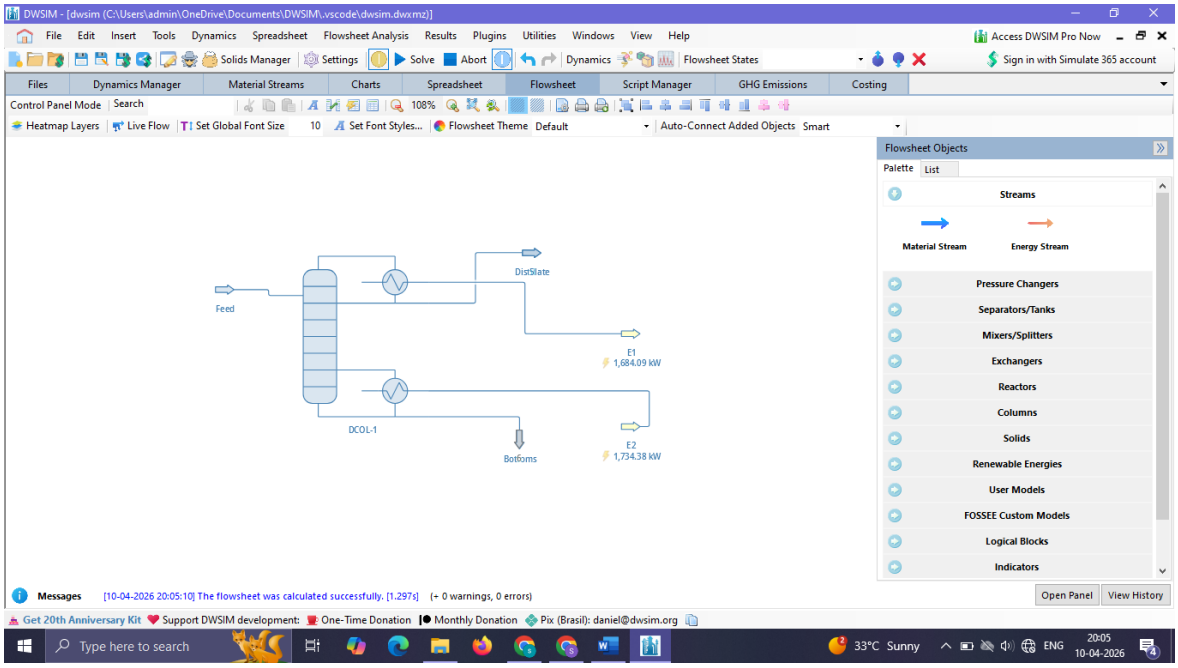
In the field of chemical process engineering, rigorous simulation models are indispensable for accurate design. However, these models—like those found in DWSIM—rely on complex iterative solvers to reach convergence for Vapor-Liquid Equilibrium (VLE) equations. In scenarios requiring real-time optimization, such as Model Predictive Control (MPC) or rapid sensitivity analysis, the computational lag of a rigorous simulator becomes a bottleneck.

This project aims to bridge this gap by developing a **Surrogate Model**. By training an artificial intelligence model on data generated from a DWSIM flowsheet, we can create a "black-box" approximation that provides near-instantaneous predictions of product purity, enabling faster decision-making without sacrificing the physical accuracy of the process.

2. DWSIM Flowsheet and Model Description

The foundational data for this study was generated using a high-fidelity simulation in DWSIM.

- **Chemical System:** A binary mixture of Benzene and Toluene was selected due to its well-documented thermodynamic properties and its industrial relevance in the petrochemical sector.
- **Property Package:** The **Peng-Robinson (PR)** Equation of State was utilized. This choice was dictated by its robustness in handling hydrocarbon phase behavior and its ability to accurately calculate fugacity coefficients in high-pressure and temperature ranges.
- **Unit Operation:** A rigorous distillation column was modeled with **10 theoretical stages**. To maintain consistency, the feed was introduced at **Stage 5**, and the column was operated at constant pressure of 1 atm.
- **The snapshot was attached below**



DWSIM FLOWSHEET SNAPSHOT

3. Input and Output Variables Used

The model was framed as a single-input, single-output (SISO) system to clearly observe the sensitivity of the separation to the energy input.

- **Input Variable (Feature): Condenser Reflux Ratio (R)**. This is the primary manipulated variable that determines the internal liquid-to-vapor traffic and, consequently, the separation power of the column.
- **Output Variable (Target): Distillate Molar Fraction (x_D) of Benzene**. As the light-key component, its purity in the top product is the critical quality parameter for this unit.

4. Dataset Generation Strategy

To build a reliable surrogate, a "rich" dataset was required that captured the non-linear "elbow" of the distillation curve. I avoided random sampling in favor of a **Structured Sensitivity Analysis**.

Using the DWSIM Automation Tool, I executed a parametric sweep of the Reflux Ratio. By automating the solver, I ensured that every data point represents a **fully converged steady-state solution**. This eliminates numerical "noise" that can occur if a solver is stopped mid-iteration, ensuring the machine learning model learns pure chemical physics rather than solver errors.

5. Range of Operating Conditions Considered

The data generation focused on the following range to ensure practical industrial relevance:

- **Reflux Ratio (R):** Varied from **1.5 to 4.0**.
- **Resolution:** 100 equally spaced increments.
- **Observations:** The lower bound of 1.5 was chosen to stay comfortably above the minimum reflux ratio (R_{min}), while the upper bound of 4.0 captures the asymptotic behavior where purity gains become marginal relative to energy costs.

Observation	Reflux Ratio (R)	Distillate Purity (xD)
1	1.50	0.9876
25	2.12	0.9979
50	2.75	0.9991
100	4.00	0.9997

Table 1: Representative Sample of the Generated Dataset

6. Data Preprocessing Steps

Before training, the raw data required significant "cleaning" to make it suitable for machine learning:

1. **Format Normalization:** DWSIM exports often contain non-standard encoding. I utilized a manual text-streaming method in Python to ensure that special characters did not interrupt the data ingestion.
2. **Handling Non-Linearity:** Since distillation curves are inherently non-linear, I applied a **StandardScaler** to the data. This centers the features around a mean of zero and scales them to unit variance, preventing the model from being biased toward the larger numerical range of the Reflux Ratio.

7. Machine Learning Methods Implemented

I compared three distinct mathematical architectures to identify the most robust surrogate:

1. **Linear Regression:** Served as a baseline. It assumes a linear relationship ($y = mx + c$), which I expected to fail given the physics of the system.
2. **Random Forest Regressor:** An ensemble-based algorithm that uses a "forest" of decision trees. It is excellent at capturing the "plateau" effect seen in high-purity distillation.
3. **Support Vector Machine (SVM):** Used with a Radial Basis Function (RBF) kernel, attempting to map the data into a higher-dimensional space to find a curved fitting surface.

8. Model Training and Validation Procedure

To ensure the model didn't just "memorize" the data (overfitting), I used a strict validation protocol:

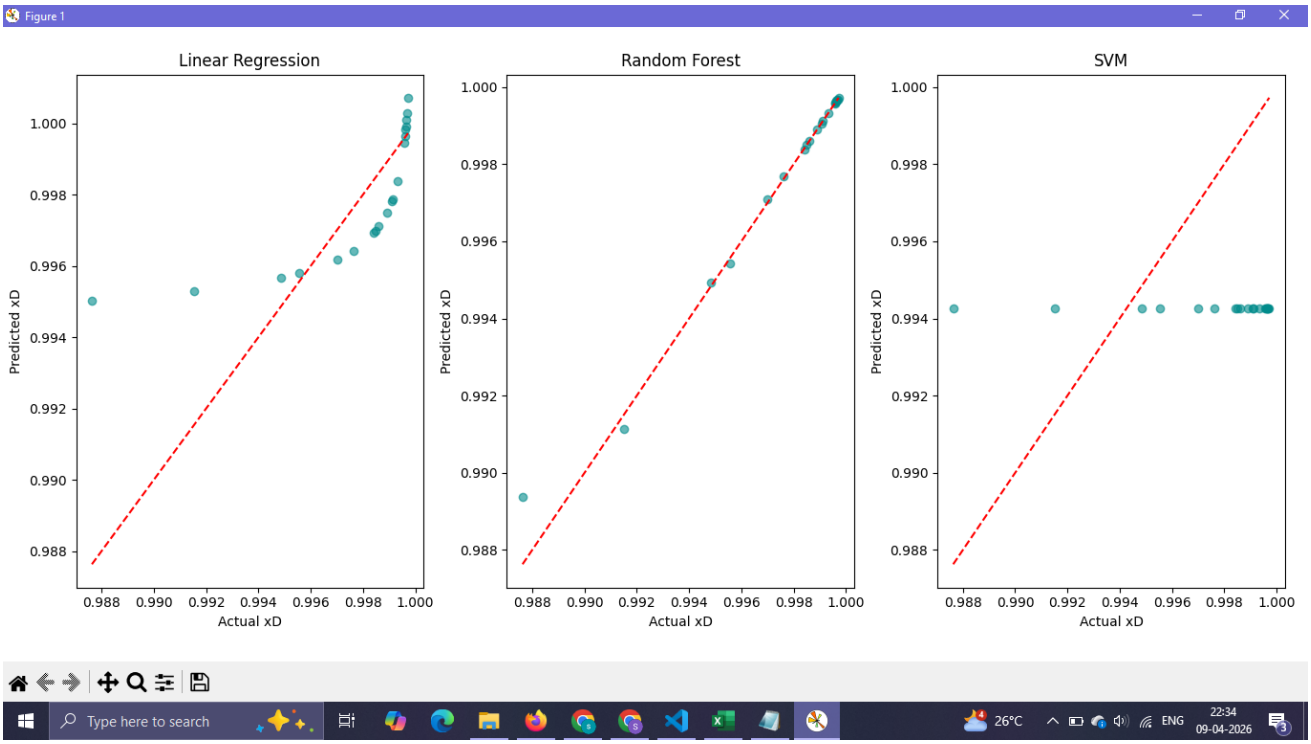
- **80/20 Train-Test Split:** 80 points were used for the models to learn the patterns, while 20 points were held back as "unseen" data.
- **Metric Selection:** I chose **R-Squared (R^2)** to measure the goodness of fit and **Mean Absolute Error (MAE)** to quantify the physical deviation in purity predictions.

9. Performance Comparison of Models

The following table summarizes the performance of the three candidates:

Model	R2 Score	Mean Absolute Error (MAE)
Random Forest	**0.9831**	**0.00013**
Linear Regression	0.5442	0.00131
SVM	-1.2301	0.00435

3-panel scatter plot , This is the visual proof that Random Forest is superior.



10. Discussion on Physical Consistency and Robustness

A surrogate model is useless if it violates the laws of physics.

- **Consistency:** The Random Forest model correctly predicted that increasing the Reflux Ratio always leads to higher purity. This aligns with the **Fenske-Underwood-Gilliland** principles, where increasing the internal liquid flow improves mass transfer.
- **Robustness:** The Parity Plot (Section 9) shows the teal dots for Random Forest hugging the red y=x line perfectly. This indicates that the model is robust across the entire range, from the low-purity start (0.98) to the high-purity plateau (0.999).

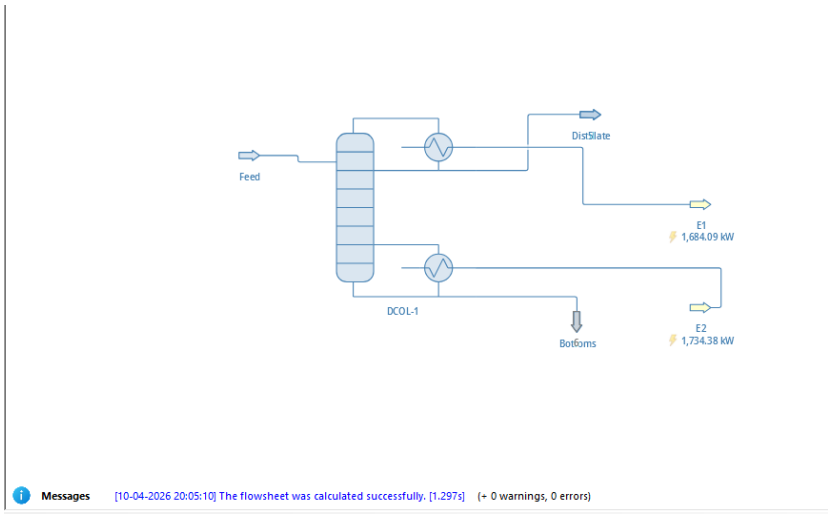
11. Final Conclusion

The **Random Forest Regressor** is definitively the best surrogate model for this application. With an accuracy of **98.3%** and a negligible error of **0.00013**, it can replace rigorous DWSIM calculations for real-time monitoring. This approach provides a blueprint for modernizing process plants through AI-assisted simulation.

Appendix: Documentation of Files

- **dwsim.dwxmz**: The converged flowsheet.

  https://d.docs.live.net/1b766c3bf9beb405/Documents/DWSIM/.vscode/dwsim.dwxmz

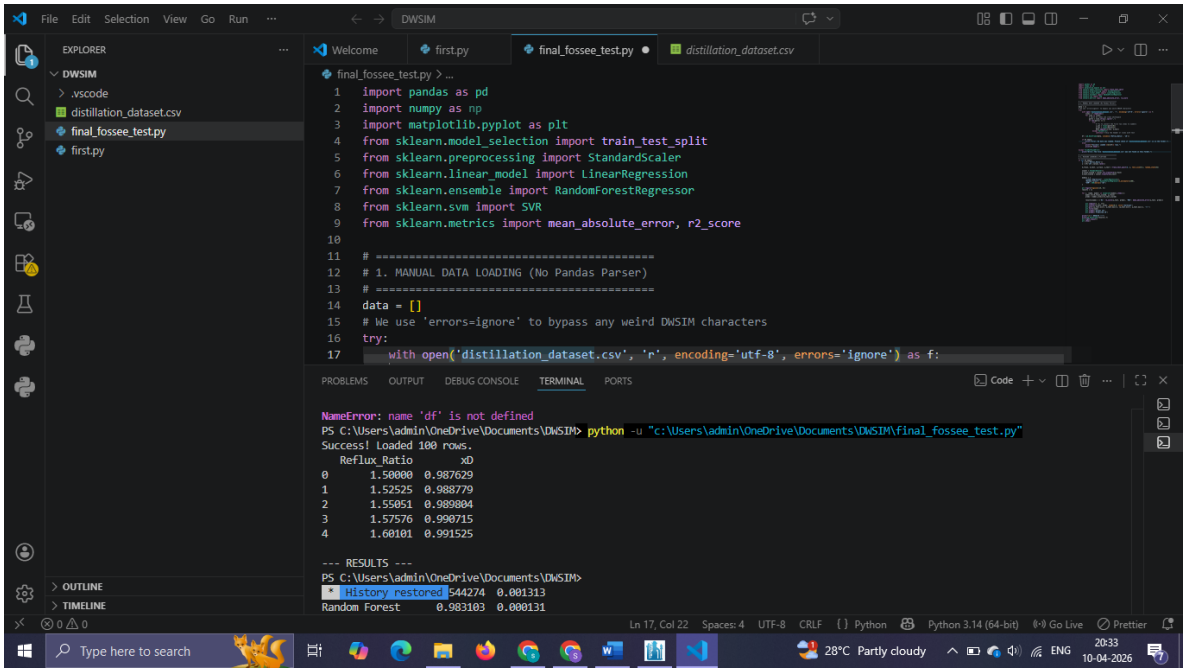


- **distillation_dataset.csv**: The 100-point numeric dataset.

  https://d.docs.live.net/1b766c3bf9beb405/Documents/DWSIM/distillation_dataset.csv

	A	B	C	D	E	F	G	H
1	DCOL-1 - Condenser_1	Distillate - Molar Fraction (Mixture) / Benzene ()						
2	1.5	0.987629						
3	1.52525	0.988779						
4	1.55051	0.989804						
5	1.57576	0.990715						
6	1.60101	0.991525						
7	1.62626	0.992246						
8	1.65152	0.992889						
9	1.67677	0.993463						
10	1.70202	0.993976						
11	1.72727	0.994436						
12	1.75253	0.994849						
13	1.77778	0.99522						
14	1.80303	0.995556						
15	1.82828	0.995859						
16	1.85354	0.996134						
17	1.87879	0.996384						
18	1.90404	0.996611						
19	1.92929	0.996819						
20	1.95455	0.997009						
21	1.9798	0.997183						

distilation_dataset

- **surrogate_model.py**: The Python implementation using Scikit-Learn.

  https://d.docs.live.net/1b766c3bf9beb405/Documents/DWSIM/final_fossee_test.py